Self-Justification Ex Machina

jpt4*

In memory of Dan E. Willard (1948–2023)

Abstract

Sebastian Oberhoff's essay *Incompleteness Ex Machina* proves Gödel's incompleteness theorems twice, once with sentences and once with programs, and concludes that Gödel's work divides into two parts: teaching arithmetic to compute, and the incompleteness theorems, which are victory laps. We look closely at the one step of the second theorem that Oberhoff takes "with some faith": that a system which can watch any run of its own proof-searcher can also prove, of the run that never happens, that it would have seen it. Willard's self-justifying axiom systems are what remains when that step is withheld. They prove every true Σ_1-class sentence of their language, they obey the first incompleteness theorem, and they take their own consistency as an axiom; Willard proved, in ordinary metamathematics, that they are consistent whenever the Π_1-class theorems of their base theory are true. We state everything at the granularity self-justification needs (language, deduction method, consistency notion, encoding), retell Willard's consistency proof as a program in Oberhoff's case format, and show that in such a system the faith step is provably equivalent to the Gödel sentence, hence unprovable. Willard's 2005 Theorem 6 then shows the faith being bought: the step is a bounded Π_1 theorem of Peano Arithmetic, and one axiom asserting that a single large number exists turns it on and the system inconsistent. Gödel's work, viewed this way, splits into three parts, not two.

> Another way to motivate the axiom system IS is to consider a philosopher's response to the question of whether multiplication is a function. His first answer would likely be "of course yes". But if one then asked the philosopher to construct a sequence of natural numbers $a_0, a_1, a_2, a_3, \ldots$ where $a_0 \geq 2$ and $a_{i+1} = (a_i)^2$, a hedge would quickly follow because a_n requires at least 2^n bits. The expected response would then be that "multiplication is a function in theory but not always in practice". — Dan Willard, SUNY Albany TR 93-10, p. 6

1 Victory Laps

Oberhoff's essay [Oberhoff 2018] does something deceptively simple. It writes out Gödel's first incompleteness theorem, Rosser's improvement of it, and the second incompleteness theorem as explicit case analyses. Then it replaces Gödel's sentence G ("I am not provable") by $\ddot{\mathsf{G}}$ = «P(P) doesn't halt», where P is the program

```
P(A) {
  for s ∈ all possible strings {
    if s is a proof in 𝓕 that A(A) doesn't halt {
      halt;
    }
```

*Revision of 25 October 2026, prepared with Claude (Anthropic) in the `sjas` research repository. Willard's results are cited from that repository's codification of his papers (`nachlass/codification/codified-sjas.md`), with page numbers referring to the author copies held there and proof statuses as recorded there.

```
  }
}
```

and observes that each case analysis can be copied almost word for word. The punchline (his §5) is that $\ddot{\mathsf{G}}$ is G wearing Clark Kent's glasses: $\mathsf{P}(\mathsf{P})$ is searching $\mathcal{F}$'s proofs for a proof of $\ddot{\mathsf{G}}$, so $\ddot{\mathsf{G}}$ says "I am not provable". His moral (§6) is a refactoring. The hard part of Gödel's work was teaching arithmetic to compute, packaged as

> **Theorem 7** (Turing Completeness of Elementary Arithmetic). *Every formal system capable of reasoning about elementary arithmetic can also reason about algorithms,*

which Oberhoff leaves "to smarter people". The incompleteness theorems themselves are victory laps.

For the first incompleteness theorem we accept this completely. This essay is about the second. Its proof in [Oberhoff 2018] contains exactly one step that Oberhoff marks as taken on trust (Theorem 6, third step):

> Now we simply observe (with some faith) that there wasn't actually anything in the reasoning we used to prove this that we couldn't carry out just as well in $\mathcal{F}$.

For the systems Oberhoff has in mind, that faith is well placed. But some formal systems can reason about algorithms in every sense the first theorem needs, and for them the faith is misplaced: they prove their own consistency, and, by a theorem of Willard's proved in ordinary metamathematics, they are consistent whenever their base theory's Π_1-class theorems are true. Dan Willard constructed them in papers from 1993 to 2020 [Willard TR 93-10, Willard 2001, Willard 2005, Willard 2006a, Willard 2020]. (We cite Willard's papers by year, as in the bibliography: "2002a" is [Willard 2002a].) This essay tells their story in Oberhoff's language. Our headline is that the faith step is not part of the victory lap. It is a third part of Gödel's work, and it has a price.

The standard of precision. Oberhoff's essay is a paper proof at the granularity the second incompleteness theorem needs: for Peano Arithmetic and its relatives, it does not matter which language, which proof system, which notion of consistency or which Gödel numbering one picks, since the system proves that the choices agree. Self-justification lives exactly where those choices stop agreeing. Willard's systems are self-justifying for one deduction method and inconsistent for another, consistent at one consistency level and not at the next, and his proofs measure Gödel numbers in bits. An exegesis that is to have the proof status of Oberhoff's must therefore fix every one of these parameters, and say where a result depends on them. Section 2 fixes them. Throughout, "Willard proved" means that the result is printed with a complete proof (status `full`); other statuses are named where they occur. Results labelled *ours* are proved here.

2 Conventions

Configurations. Following Willard, a *configuration* (α, D) is an axiom system α together with a deduction method D; $\alpha \vdash_D \varphi$ means that φ has a D-proof from α. D is *α-sound* if $\alpha \vdash_D \varphi$ implies that φ holds in every model of α, and *α-complete* if the converse holds (2001, §7, p. 37). All deduction methods in this essay (Hilbert-style deduction, semantic tableaux, and Willard's tableau lists below) are sound and complete in this sense. They differ in proof *length*, and self-justification is a phenomenon of length.

Fact A (Willard 2001, Lemma 7.1B, p. 37, `full`)**.** *If D is α-sound and α-complete, $\alpha \vdash_D \varphi$, $\alpha \vdash_D \psi$, and $\varphi \wedge \psi \supset \chi$ holds in every model of α, then $\alpha \vdash_D \chi$.*

So the theorems of (α, D) are closed under logical consequence. What is not closed is the size of proofs: for a cut-free D, Willard notes, the combined proof "could then correspond to a bit string more than exponentially larger than the combined lengths of the three initial proofs" (2001, p. 38, text layer).

Languages. Three languages occur.

- The language of arithmetic, with 0, successor, + and ×, and the usual classes Δ_0, Π_1, Σ_1.
- Willard's 2001 language: the seven *grounding functions* (truncated subtraction, integer division, predecessor, maximum, logarithm, root, and *Count*, all *non-growth*: $F(\vec{a}) \leq \max(\vec{a})$) and one constant symbol $\bar{n}$ for every natural number n (2001, pp. 9–10). Its classes are written Δ_0^-, Π_1^-, Σ_1^-. Addition is not a function symbol; Group-1 of IS(A) asserts its totality by the Π_2^- axiom $\forall x\, \forall y\, \exists z\ z - x = y$ (2001, Eq. 6).
- Willard's 2005 *U-Grounding* language: six grounding functions plus the growth functions + and $Double(x) = x + x$, eight function symbols in all, with constants $\bar{c}_0$, $\bar{c}_1$ for 0 and 1 (2005, §2, p. 4, and §3, p. 8). Every $n \geq 2$ is named by a term $\underline{n}$ with at most $2 \log_2 n$ occurrences of + and *Double* applied to 1. Its classes are Δ_0^*, Π_1^*, Σ_1^*. Multiplication is not a function symbol; its graph $M(x, y, z)$ is a Δ_0^* formula (2005, Eq. 4), so the system knows multiplication's algebra but not its totality. Multiplication by any fixed constant is total, through iterated addition; what is withheld is the *uniform* operation $x \cdot y$ (codified statement §3.1).

We write "bounded", "Π_1-class" and "Σ_1-class" for whichever of these classes is in force.

Consistency notions. For a configuration (α, D), Willard distinguishes (2005, p. 5): *Level(0−)*: there is no D-proof of $0 = 1$ from α; *Level(1)*: there is no Π_1^* sentence Ψ such that both Ψ and $\neg\Psi$ have D-proofs from α. "All the definitions of consistency, from Level(0-) up to Level(n) for any n, are equivalent to each other under strong enough models of arithmetic. However, many weak axiom systems do not have a mathematical strength to recognize this equivalence" (2005, p. 6, text layer). Willard's rule for choosing among them: "it is desirable to use the weakest possible definition ... when one is seeking to generalize the Second Incompleteness Theorem. On the other hand, the opposite is true when one seeks to develop 'boundary-case exceptions' " (2002c, p. 2, text layer). This is why his negative results below use Level(0−) and his positive ones Level(1).

Provability, halting, and the Gödel sentence. For each configuration in this essay, $\mathrm{Prf}_\alpha^D(x, y)$ ("y is the Gödel number of a D-proof from α of the sentence with Gödel number x") is a bounded formula, and $\mathrm{Prov}_\alpha^D(x) := \exists y\, \mathrm{Prf}_\alpha^D(x, y)$. Oberhoff's «$\mathsf{A}(\mathsf{A})$ halts», for the proof-searching programs P, B, C of this essay, is formalized as "the search succeeds": an existential statement about Prf, not about execution traces. The sentence $\ddot{\mathsf{G}}$ is Willard's fixed point (2001, Eq. 63, p. 38):

$$\Gamma(g) := \forall x\, \forall h \leq x\, [\mathrm{SUBST}(g, h) \supset \neg \mathrm{Prf}_\alpha^D(h, x)], \qquad \ddot{\mathsf{G}} := \Gamma(\bar{n}),$$

where $\mathrm{SUBST}(g, h)$ says that h is the code of the sentence obtained by substituting the numeral of g into the formula coded by g, and n is the Gödel number of $\Gamma(g)$. $\ddot{\mathsf{G}}$ is a Π_1-class sentence, and $\neg\ddot{\mathsf{G}} = \exists x\, \exists h \leq x\, [\mathrm{SUBST}(\bar{n}, h) \wedge \mathrm{Prf}_\alpha^D(h, x)]$ is «$\mathsf{P}(\mathsf{P})$ halts». Let $c := \ulcorner \ddot{\mathsf{G}} \urcorner$.

Fact B (Fixed point; Willard 2001, proof of Theorem 7.2, p. 39)**.** *Suppose that α proves every true bounded sentence, proves that $<$ is a total order, and proves the uniqueness sentence* (*U*) $\forall h\,[\mathrm{SUBST}(\bar{n}, h) \supset h = \bar{c}]$. *Suppose also that the encoding satisfies* (*E*)*: the Gödel number of a D-proof always exceeds the Gödel number of the sentence it proves. Then*

$$\alpha \vdash_D \ddot{\mathsf{G}} \leftrightarrow \forall y\, \neg \mathrm{Prf}^D_\alpha(\bar{c}, y). \tag{FP}$$

Proof. ($\Leftarrow$) Given x and $h \leq x$ with $\mathrm{SUBST}(\bar{n}, h)$, (U) gives $h = \bar{c}$, and the hypothesis gives $\neg\mathrm{Prf}(\bar{c}, x)$. ($\Rightarrow$) Given y, either $y < \bar{c}$ or $\bar{c} \leq y$. In the first case $\neg\mathrm{Prf}(\bar{c}, y)$ follows from the bounded sentence $\forall y < \bar{c}\, \neg\mathrm{Prf}(\bar{c}, y)$, which is true by (E) and therefore provable. In the second, instantiate $\ddot{\mathsf{G}}$ at $x = y$, $h = \bar{c}$, using the true bounded sentence $\mathrm{SUBST}(\bar{n}, \bar{c})$. Fact A turns these semantic consequences into D-proofs. □

Willard's text gives the key point, (U): "$\ulcorner\Theta\urcorner$ is the unique integer satisfying $\mathrm{SUBST}(\bar{n}, \ulcorner\Theta\urcorner)$ ... it is provable from PAX as a Π_1^- theorem" (2001, p. 39, text layer), so (U) holds in any system that inherits the Π_1^- theorems of PAX, Peano Arithmetic extended by the grounding functions. The order and (E) are our explicit additions. (E) is an encoding hypothesis; we checked it against Willard's encodings (2005, Appendix A, pp. 39–40; 2001, Appendix B, p. 51, the same scheme with curly brackets): a tableau proof is a base-64 string that opens with a bracket byte and then stores the root $\neg\Phi$ in prenex form, at least one byte longer than Φ, and every symbol byte has leading bit 1, so the proof's value exceeds $64^{|\Phi|} > \ulcorner\Phi\urcorner$. Tableau lists, which concatenate tableau proofs, inherit (E) from their last component.

Relative consistency. Every consistency result in this essay, Willard's and ours, is a theorem of ordinary metamathematics proved under a hypothesis, typically that the Π_1-class theorems of a base theory A are true. That hypothesis is not decidable, and no system here is "proved consistent" outright. We have not determined how much of the metatheory Willard's proofs use.

3 Where the Faith Lives

Here is the first half of Oberhoff's proof that $\ddot{\mathsf{G}}$ is undecidable (his §4, a second proof of his Theorem 4):

Suppose $\ddot{\mathsf{G}}$ is provable:
Or $\neg\ddot{\mathsf{G}}$ isn't provable:

- ⇒ Because $\ddot{\mathsf{G}}$ is provable $\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$.
- ⇒ Because this sequence leads to a terminal state this gives a proof of «$\mathsf{P}(\mathsf{P})$ halts» = $\neg\ddot{\mathsf{G}}$. ↯

Theorem 6 asks $\mathcal{F}$ to run this argument on itself. Fix a configuration $(\mathcal{F}, D)$ and write Prov for $\mathrm{Prov}^D_{\mathcal{F}}$. As statements that $\mathcal{F}$ must prove, the argument has three steps.

(1) $\neg\ddot{\mathsf{G}} \supset \mathrm{Prov}(\bar{c})$: «if $\mathsf{P}(\mathsf{P})$ halts, then $\ddot{\mathsf{G}}$ is provable». This is one half of (FP). It is not literally trivial: $\neg\ddot{\mathsf{G}}$ speaks of "the sentence h with $\mathrm{SUBST}(\bar{n}, h)$", and identifying h with $\ddot{\mathsf{G}}$ needs the uniqueness sentence (U). With (U), it is cheap.

(2) $\mathrm{Prov}(\bar{c}) \supset \mathrm{Prov}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner)$: «if $\ddot{\mathsf{G}}$ is provable, then $\neg\ddot{\mathsf{G}}$ is provable». This is the step "$\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$". Inside $\mathcal{F}$ it reads: *for every proof s of $\ddot{\mathsf{G}}$ there is a proof of «$\mathsf{P}(\mathsf{P})$ halts»*, namely the proof that exhibits s and checks it.

(3) $\mathrm{Prov}(\bar{c}) \wedge \mathrm{Prov}(\ulcorner \neg \ddot{\mathsf{G}} \urcorner) \supset \mathit{Inc}$, where Inc is the negation of the consistency statement in use. For Level(1), whose statement is $\forall x \forall y \forall p \forall q\ \neg[\mathrm{Pair}(x,y) \wedge \mathrm{Prf}(x,p) \wedge \mathrm{Prf}(y,q)]$ with $\mathrm{Pair}(x,y)$ the bounded formula "x codes a Π_1-class sentence and y its negation" (2005, Eq. 6), step (3) is an instance, given the true bounded sentence $\mathrm{Pair}(\bar{c}, \ulcorner \neg \ddot{\mathsf{G}} \urcorner)$. For Level(0−) it is a formalized combination of two proofs into a proof of $0 = 1$, which for a cut-free D is the expensive construction Fact A warns about.

Chaining (1)–(3) gives $\mathcal{F} \vdash_D \neg \ddot{\mathsf{G}} \supset \mathit{Inc}$, that is, $\mathcal{F} \vdash_D \neg \mathit{Inc} \supset \ddot{\mathsf{G}}$: «$\mathcal{F}$ is consistent $\Rightarrow \ddot{\mathsf{G}}$», which is what Theorem 6 needs.

The only step that asks $\mathcal{F}$ to know something substantial about itself is (2). In the first incompleteness theorem, "which we can write out in $\mathcal{F}$" was a fact *about* $\mathcal{F}$, established by us from outside: we are handed a halting run and we produce the proof. In the second theorem it has to be a theorem *of* $\mathcal{F}$, stated for every s at once. The two deserve separate names.

Definition C (Watching)**.** $(\mathcal{F}, D)$ *watches* if $\mathcal{F} \vdash_D \sigma$ for every true Σ_1-class sentence σ of its language.

Definition D (Knowing that one watches)**.** $(\mathcal{F}, D)$ *knows that it watches* at a Σ_1-class sentence σ if $\mathcal{F} \vdash_D \sigma \supset \mathrm{Prov}(\ulcorner \sigma \urcorner)$, and *knows that it watches* if it does so at every Σ_1-class sentence.

A program halts if and only if a Σ_1 sentence about it is true, so watching is external Σ_1-completeness, and knowing that one watches is formalized Σ_1-completeness. The third Hilbert–Bernays–Löb derivability condition [Hilbert–Bernays 1939, Löb 1955], $\mathrm{Prov}(\ulcorner \varphi \urcorner) \supset \mathrm{Prov}(\ulcorner \mathrm{Prov}(\ulcorner \varphi \urcorner) \urcorner)$, is the instance at the Σ_1 sentences $\sigma = \mathrm{Prov}(\ulcorner \varphi \urcorner)$, the halting statements of proof searches.

Lemma E (ours)**.** *Let $(\mathcal{F}, D)$ watch, and let (FP) hold.*

(a) $(\mathcal{F}, D)$ knows that it watches at every true *Σ_1-class sentence.*

(b) Step (2) is $\mathcal{F}$-provably equivalent to knowing that one watches at $\neg \ddot{\mathsf{G}}$: $\mathcal{F} \vdash_D$ (2) $\leftrightarrow$ $[\neg \ddot{\mathsf{G}} \supset \mathrm{Prov}(\ulcorner \neg \ddot{\mathsf{G}} \urcorner)]$.

Proof. (a) If σ is true, $\mathcal{F} \vdash_D \sigma$ by watching, so some D-proof m of σ exists; $\mathrm{Prf}(\ulcorner \sigma \urcorner, \bar{m})$ is a true bounded sentence, so $\mathcal{F}$ proves it and hence $\mathrm{Prov}(\ulcorner \sigma \urcorner)$, and the implication follows by Fact A. (b) By (FP), $\mathcal{F} \vdash_D \neg \ddot{\mathsf{G}} \leftrightarrow \mathrm{Prov}(\bar{c})$, and the two implications differ only by this replacement in the antecedent; apply Fact A. □

So the faith step is not a claim about runs that happen. Those a watching system sees, and Lemma E(a) gives it the knowledge for free. The faith is a claim about a run that does not happen: if $\ddot{\mathsf{G}}$ is true, $\neg \ddot{\mathsf{G}}$ is a false Σ_1 sentence, and step (2) asserts that *had* $\mathsf{P}(\mathsf{P})$ halted, $\mathcal{F}$ would have seen it.

Here is what each result of [Oberhoff 2018] uses:

Result in [Oberhoff 2018]	Assumes about $\mathcal{F}$	Watches	Knows it watches
Thm 4 (G1 via halting)	honest; complete (for contradiction)	—	—
§4, $\ddot{\mathsf{G}}$ undecidable	consistent; honest in case 2	case 1	—
Thm 5 (Rosser, via B)	consistent	both cases	—
Thm 6 (G2)	consistent	yes	step (2)

Theorem 4 needs only that $\mathcal{F}$ can *state* halting facts honestly; the search does the rest. Case 2 of the §4 argument uses only that $\mathcal{F}$'s Σ_1 theorems are true. Rosser's case 2 also uses a Π_1 fact, that a

run which has entered `while true {}` never leaves it; Oberhoff's footnote 12: "Any formal system capable of reasoning about algorithms can prove that 'while true {}' runs forever; otherwise it ought to be ashamed of itself." Apart from that, every result in the essay except the second incompleteness theorem uses watching and nothing more. The classical proof of the second theorem runs through the derivability conditions; the computational retelling shows what the third condition *says*.

4 Transcripts Are Bigger Than Runs

Why should knowing be harder than watching? The natural proof of step (2) takes a proof s of $\ddot{\mathsf{G}}$ and produces the proof t of «$\mathsf{P}(\mathsf{P})$ halts» that exhibits s and checks it. In a language whose only growth functions are $+$ and *Double*, exhibiting s costs a term, and terms are expensive.

Lemma F (ours)**.** *In Willard's 2005 language and encoding, every semantic tableau proof that contains a closed term of value $v \geq 2$ has Gödel number greater than v^5.*

Proof. A closed term with L function symbols built from $\bar{c}_0$, $\bar{c}_1$ has value at most 2^L: non-growth functions do not increase the maximum of their arguments, and $+$ and *Double* at most double it (this is the bound Willard uses in 2005, Lemma 1, case (c), p. 24). So a term of value v has $L \geq \log_2 v$ symbols, and the node containing it has U-Depth at least L. Willard's Eq. (20) (2005, p. 26) states that his encoding satisfies U-Height$(p) < \frac{1}{5}\log_2 p$ for every proof p. Hence $\log_2 p > 5L \geq 5\log_2 v$. □

So the transcript t of step (2) is bigger than the fifth power of the run s it records, and the natural proof of step (2) asks $\mathcal{F}$ to believe that $s \mapsto t$ is defined everywhere. A totality claim for a function that grows like a power is a claim about multiplication, which these systems do not make. This is a heuristic, not a theorem: step (2) is vacuously true when $\ddot{\mathsf{G}}$ is, and a system could in principle prove it without exhibiting anything. Section 7 closes that loophole by a different route, and Section 8 shows the heuristic as an exact theorem of Willard's.

Willard's first statement of the idea continues this essay's epigraph (TR 93-10, p. 7, verbatim):

> This means there will always be adequate memory for writing b_n's binary encoding when there is sufficient memory for storing a semantic tableaux proof of its existence (whereas the same is plainly not true *when multiplication is assumed to be a function*).
>
> For instance, multiplication allows one to establish the existence of a number a_{300} (in a proof with approximately 300 lines), whose binary encoding requires *more digits* than the number of atoms in the universe.

Here $b_{i+1} = b_i + b_i$ and $a_{i+1} = (a_i)^2$. Starting both at 2, $b_n = 2^{n+1}$ and $a_n = 2^{2^n}$: a proof of n doublings names a number of $n+1$ bits, and a proof of n squarings names one of 2^n bits. Willard's later statement compares the Gödel number of a proof with the integer it shows to exist (2006a, Remark 3, p. 24, verbatim):

> In particular, every proof of the existence of an integer N using the additive convention will have a larger Gödel number than N. In contrast, the multiplicative convention allows for the Gödel number of such proofs to have a sharply smaller magnitude than N (when for instance $N = 2^{2^k}$ and k is larger than some fixed constant). In essence, this difference in magnitude is the reason the Second Incompleteness Theorem will generalize under the multiplicative naming paradigm — while Section 4 showed boundary-case exceptions to it exist under the additive naming convention.

5 A Machine That Says "I Am Consistent"

Oberhoff's moral for the second theorem is "if you declare yourself a very stable genius, you're not". Let us build the genius with his tools. Fix a base system $\mathcal{F}_0$, a deduction method D, and a consistency level, and consider

```
C(A) {
  for s ∈ all possible strings {
    if s is a contradiction in F_0 + «A(A) doesn't halt» {
      halt;
    }
  }
}
```

where a *contradiction* is a D-proof of $0 = 1$ (Level(0−)) or a pair of D-proofs of a Π_1-class sentence and of its negation (Level(1)). (We picked C because it is a circle that never quite closes; §6 explains the joke.) Let

$$\ddot{\mathsf{W}} = \text{«}\mathsf{C}(\mathsf{C}) \text{ doesn't halt»}, \qquad \mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}.$$

Just as $\ddot{\mathsf{G}}$ says "I am not provable", $\ddot{\mathsf{W}}$ says that $\mathcal{F}$ is consistent at the chosen level. So $\mathcal{F}$ proves its own consistency in one line: the claim is an axiom.

C is not an analogy. It is Willard's construction read as a program. In 2005 (pp. 9–10), $ExPrf^{D}_{\mathrm{UNION}(A)}(h,t,p)$ says that p is a D-proof of t from Groups 0–2 plus the sentence coded h, and $SubstPrf^{D}_{\mathrm{UNION}(A)}(g,t,p)$, his Eq. (7), combines it with $\mathrm{SUBST}(g,h)$: the input A is g, and «$\mathsf{A}(\mathsf{A})$ doesn't halt» is the sentence h with $\mathrm{SUBST}(g,h)$. His Group-3 axiom, Eq. (8), is $\Gamma(\bar{n})$ for

$$\Gamma(g) := \forall x\, \forall y\, \forall p\, \forall q\, \neg\big[\mathrm{Pair}(x,y) \wedge SubstPrf^{D}_{\mathrm{UNION}(A)}(g,x,p) \wedge SubstPrf^{D}_{\mathrm{UNION}(A)}(g,y,q)\big],$$

which is «$\mathsf{C}(\mathsf{C})$ doesn't halt» at Level(1); 2001's Appendix B does the same at Level(0−). Willard (2016, Definition 3.4 and Example 3.5) remarks that constructing such a sentence is "relatively easy, via an application of the Fixed Point Theorem, but this sentence is ironically typically useless!"

Proposition G (The genius falls; standard)**.** *Let $\mathcal{F}_0$ be a recursively axiomatized extension of $I\Sigma_1$ in the language of arithmetic, D Hilbert-style deduction,* Prov *the canonical Σ_1 provability predicate, and $\ddot{\mathsf{W}}$ as above at either level. Then $\mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}$ is inconsistent, and $\mathsf{C}(\mathsf{C})$ halts.*

Proof. Suppose $\mathcal{F}$ is consistent.

- ⇒ $\mathcal{F}$ proves its own consistency: $\ddot{\mathsf{W}}$ is one of its axioms, $I\Sigma_1$ proves the fixed-point equivalence between $\ddot{\mathsf{W}}$ and the canonical consistency statement of $\mathcal{F}_0 + \ddot{\mathsf{W}}$ (the diagonal lemma), and $I\Sigma_1$ proves the two consistency levels equivalent for $\mathcal{F}$.
- ⇒ $\mathcal{F}$ is a recursively axiomatized extension of $I\Sigma_1$, so the derivability conditions hold for its canonical provability predicate, provably in $I\Sigma_1$ [Hájek–Pudlák 1993]. In Oberhoff's terms, $\mathcal{F}$ knows that it watches, and steps (1)–(3) are theorems of $\mathcal{F}$.
- ⇒ So by the second incompleteness theorem (Oberhoff's Theorem 6, with its faith step now a theorem), $\mathcal{F}$ cannot prove its own consistency. ↯

So $\mathcal{F}$ is inconsistent. A contradiction from $\mathcal{F}$ is then a finite object that exists, and $\mathsf{C}(\mathsf{C})$ finds it. □

Willard records that "Kleene, Rogers and Jeroslow [...] each warned their readers that most axiom systems, similar to α^*, were useless on account of their inconsistency, although they were

technically well-defined" (2005, §1). Two of his own theorems are instances in his own languages. XIS^{λ}(PAX), his tableau system with multiplication total and a reflection-style Group-3 axiom, "is inconsistent" for every λ (2001, Theorem 7.3, p. 40, `full`). And Section 8 describes the finest instance, where the system falls because of one added axiom saying that one number exists.

6 Willard's Machine

Willard keeps C and $\ddot{\mathsf{W}}$ and changes $\mathcal{F}_0$ and D. We describe the 2005 system $\mathrm{IS}_D(A)$ and note where the 2001 system $\mathrm{IS}(A)$ differs.

The system $\mathrm{IS}_D(A)$ (2005, §3, pp. 8–10). A is any axiom system in the U-Grounding language whose proof predicate Prf_A is Δ_0^*, and D is a deduction method.

- **Group-0** (names): four axioms fixing $\bar{c}_0$, $\bar{c}_1$, $+$ and *Double*, so that every $n \geq 2$ has the name $\underline{n}$ of $O(\log n)$ symbols.
- **Group-1** (watching): a finite set of true Π_1^* sentences which, with Group-0, proves every Δ_0^* sentence true in the standard model. Willard's Table I (2001, pp. 63–64) is one such set.
- **Group-2** (inheritance): for each Π_1^* sentence Φ, the axiom $\forall p\,\{\mathrm{Prf}_A(\ulcorner\Phi\urcorner, p) \rightarrow \Phi\}$ (2005, Eq. 5). The schema is indexed by all Π_1^* sentences, not by A's theorems; for Φ that A does not prove, the axiom is vacuously true.
- **Group-3** (self-reference): the single Π_1^* sentence $\Gamma(\bar{n})$ of §5, Level(1) consistency of $\mathrm{IS}_D(A)$ under D, with $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}(t,p)$ encoded as $\mathit{SubstPrf}^D_{\mathrm{UNION}(A)}(\bar{n},t,p)$. Willard stresses that the Π_1^* form is essential (2005, p. 10).

The deduction methods are *semantic tableaux* (2005, §2, pp. 6–7: a tree rooted at $\neg\Phi$ in prenex form, with elimination rules only and no cut, every branch closed by some Υ and $\neg\Upsilon$) and Tab$-U_1^*-$List: a list of tableau proofs, each of which may use the earlier conclusions as axioms, all intermediate conclusions being Π_1^* or Σ_1^* (2005, p. 7). The latter is a restricted modus ponens: lemmas are allowed, provided each has at most one unbounded quantifier block.

$\mathrm{IS}(A)$ of 2001 differs in three respects: its language is the 2001 grounding language, with a constant for every number and addition total by an axiom rather than a function symbol; D is semantic tableaux; and its Group-3 axiom asserts Level$(0-)$ consistency, $\forall y\,\neg \mathit{SemPrf}_{\mathrm{IS}(A)}(\ulcorner 0 = 1\urcorner, y)$ (2001, Eq. 8), with a Π_1^- encoding (2001, Theorem C.3, p. 60). Willard assumes throughout that A extends Peano Arithmetic (2001, p. 9).

Theorem H (Willard)**.**

(a) (2005, Theorems 3 and 5, pp. 27, 29, `full`.) If every Π_1^* theorem of A is true in the standard model, then $\mathrm{IS}_D(A)$ is consistent, both for D = semantic tableaux and for $D =$ Tab$-U_1^*-$List.

(b) (2001, Theorem 4.3, p. 20, `full`.) If A is regularly consistent*, meaning that Prf_A has a Δ_0^- encoding and every Group-2 axiom of $\mathrm{IS}(A)$ is true (2001, p. 12), then $\mathrm{IS}(A)$ is consistent.*

Both are relative consistency theorems in the sense of §2. Willard also states that IS's construction fails for Hilbert deduction: by Pudlák and Solovay, "if D denotes Hilbert deduction then our $\mathrm{IS}_D(A)$ system will be automatically inconsistent" (2005, p. 8, text layer).

Lemma I (ours)**.** *For D = semantic tableaux or* Tab$-U_1^*-$List*, $(\mathrm{IS}_D(A), D)$ watches.*

Proof. Let $\sigma = \exists \vec{x}\, \delta(\vec{x})$ be a true Σ_1^* sentence with witnesses $\vec{k}$. Then $\delta(\underline{\vec{k}})$ is a true Δ_0^* sentence, which Groups 0 and 1 prove, and σ is its logical consequence. Apply Fact A. □

The same holds for IS(A) with $A \supseteq$ PAX: a system that verifies the Π_1^- theorems of PAX "obviously has a capacity to prove every Δ_0^- sentence which is true" (2001, p. 39, text layer), and IS(A) names every number by a constant. For general programs, Willard notes that the predicate *Turing*(x, y, z) is expressible in IS(A) (TR 93-10, p. 6), so halting of any program is a Σ_1^- sentence there; the arguments below need only the proof-searchers, whose halting is formalized through Prf directly.

Lemma J (ours)**.** *If every Π_1^* theorem of A is true, then every theorem of* IS$_D(A)$ *is true in the standard model.*

Proof. Groups 0 and 1 are true (Willard: "It is obvious that all IS$_D(A)$'s Group-0 and 1 axioms are valid under the standard model", 2005, p. 20, text layer). Group-2 is true by hypothesis. Group-3 is true because IS$_D(A)$ is consistent (Theorem H(a)) and D is sound. Theorems of true axioms under a sound D are true. □

So IS$_D(A)$ is honest in Oberhoff's sense, under the hypothesis of Theorem H. It obeys the first incompleteness theorem: by Oberhoff's §4 argument (spelled out as Proposition K(c) below), $\ddot{\mathsf{G}}$ is unprovable, and $\neg\ddot{\mathsf{G}}$ is false, hence unprovable by honesty. Rosser's version also goes through for $A \supseteq$ PA+ (Peano Arithmetic with the U-Grounding function symbols, 2005, p. 32). In both cases the finite search that Oberhoff writes out is a true bounded fact, which Groups 0 and 1 prove; in case 2, $\mathsf{B}(\mathsf{B})$ enters its loop at a definite step, Peano Arithmetic then proves the Π_1^* sentence «$\mathsf{B}(\mathsf{B})$ doesn't halt», and Group-2 hands it to IS$_D(A)$. Willard said so in his first technical report: IS(A) is "a class of structures, lying halfway between Peano Arithmetic and Presburger Arithmetic, which satisfy Godel's First but not Second Incompleteness Theorem" (TR 93-10, p. 6, verbatim).

Why it is consistent, in Oberhoff's format. Willard's 2005 proof needs two definitions. A Π_1^* sentence $\forall \vec{x}\, \psi$ is *App*$\forall(a)$*-valid* if ψ holds for all $\vec{x} \le a$; a Σ_1^* sentence $\exists \vec{x}\, \psi$ is *App*$\exists(b)$*-valid* if it has a witness $\vec{x} \le b$ (Eqs. 16–17; bounded quantifiers are left alone). For $a > b \ge 1$, α is *Normed*(a, b) if every axiom is a Π_1^* sentence with App$\forall(a)$ validity or a Σ_1^* sentence with App$\exists(b)$ validity, in the U-Grounding language with no other function symbol (Definition 4, p. 19). D is *θ-compactified*, for a constant $0 < \theta < 1/3$, if whenever $p \le \lceil a/b \rceil + 1$ is a D-proof of Φ from a Normed(a, b) system, a Σ_1^* Φ is App$\exists(b \cdot p^\theta)$-valid and a Π_1^* Φ is App$\forall(a/p^\theta)$-valid (Definition 5, p. 19). In words: a proof p can push an existential witness out by at most a factor p^θ, and can shrink a universal's certified range by at most the same factor. Tableaux are $\frac{1}{4}$-compactified (Theorem 2, p. 26, `full`), and so are tableau lists (Theorem 4, p. 28, `full`). Here is Theorem 1 (p. 19, `full`), with $\theta = \frac{1}{4}$:

Suppose IS$_D(A)$ **is inconsistent:**

- ⇒ Then $\mathsf{C}(\mathsf{C})$ halts: there are D-proofs j_1 of a Π_1^* sentence Φ and j_2 of $\neg\Phi$. Choose them with $m = \max(j_1, j_2)$ as small as possible.
- ⇒ Groups 0–2 are true. Group-3 is false, but its first counterexample is a pair of proofs whose maximum is m, so it is App$\forall(m-1)$-valid. Hence IS$_D(A)$ is Normed$(m-1, 1)$, which presupposes $m - 1 > 1$; under Willard's encoding every proof has Gödel number far above 3.
- ⇒ The proof $j_1 \le m$ certifies the universal Φ out to $(m-1)/m^{1/4}$. The proof $j_2 \le m$ finds a witness for the existential $\neg\Phi$ at most $m^{1/4}$.
- ⇒ Since $(m-1)/m^{1/4} > m^{1/4}$, the witness for $\neg\Phi$ lies where Φ has been certified. ↯

The work is in the third arrow, θ-compactification, and it is a program. For Part A of Theorem 2 (p. 26), suppose a tableau p proves a Σ_1^* sentence Φ from a Normed(a, b) system but Φ has no witness $\leq b\,p^{1/4}$:

Suppose p is a tableau proof of Φ and Φ is not App$\exists(b\,p^{1/4})$-valid:

- $\Rightarrow$ Then the root $\neg\Phi$ is App$\forall(b\,p^{1/4})$-valid, and the axioms together with the root form a Normed$(b\,p^{1/4}, b)$ system.
- $\Rightarrow$ Run *Probe* (Lemma 1, p. 23, `full`): walk down p from the root; at a fork take a child that still satisfies the branch condition (each universal valid up to $b\,p^{1/4}$, each existential witnessed below b, each bounded sentence true) under the current valuation; when an existential is eliminated, give its new parameter the least value that keeps the branch true. Every parameter introduced at a node of U-Depth Δ gets a value at most $b \cdot 2^{\Delta}$ (Eq. 19), because terms grow at most by doubling.
- $\Rightarrow$ The walk succeeds if the tree's U-Height is below $\log_2(b\,p^{1/4}) - \log_2 b = \frac{1}{4}\log_2 p$, which Eq. (20) guarantees.
- $\Rightarrow$ A branch found by *Probe* cannot close (Lemma 2, p. 25, `full`): a closing pair must be bounded sentences (footnote 3), and both would be true under the valuation.
- $\Rightarrow$ So p has an open branch, and is not a proof. $\lightning$

Part B, for Π_1^* sentences, permutes the quantifiers (Appendix B, pp. 40–41). Theorem 3, for tableaux, is then immediate, and Theorem 5, for tableau lists, follows the same way from Theorem 4.

Willard's earlier intuition for the 2001 system, his "Assertion ++", is the same horizon argument seen from the other side (2001, Remark 4.4, p. 20, verbatim): "any proof y of $0 = 1$ from IS(A) will be automatically associated with an ordered pair (p, u) where $p \leq \text{VALUE}(u) < y$ and p represents ANOTHER proof of $0 = 1$". He calls this a *vacuous truth*, an implication whose hypothesis is never satisfied. The 2001 proof replaces it by an eleven-case induction showing that the least tableau proof y of $0 = 1$ contains a branch on which every sentence is true once its unbounded quantifiers are cut off at $y - 1$ (Lemma 4.8, p. 23, `full`), which cannot close (Lemma 4.6, p. 22, `full`). The Group-3 axiom is false, but its cut-off at $y - 1$ is true, because y is the least refutation. The naming bounds are Lemma 4.1 (a constant mentioned in y has value below $\sqrt{y}$, p. 17, `full`) and Lemma 4.2 (along such a branch every parameter has value below y, p. 19, `sketch`), whose intuition is that fewer than $\frac{1}{3}\log_2 y$ doublings take values at most to $y^{1/2} \cdot y^{1/3} = y^{5/6} < y$ (the product is our unpacking of the sketch). The formal proof uses a different majorant, Part II of the affirmative-branch definition, $\text{VALUE}(u) < (k + \sqrt{y}) \cdot 2^{\lfloor \beta_i/6 \rfloor - 2}$ (Eq. 26, p. 22), where β_i is the bit-length of the first i levels of the branch and each doubling is paid for by the six bits of Case 11. Willard later found this proof "hard to visualize" and replaced it by Theorem 1 (2005, Remark 2, p. 21).

Three remarks.

The proof is a program. Probe takes any purported proof and returns an open branch together with values for its parameters. Willard's consistency proof is therefore a machine that defeats each refutation by running it, in a universe barely larger than the refutation itself. That is self-justification ex machina in the most literal sense. (And it is why we named the searcher C: it hunts for a tree all of whose branches close, and the walk always finds one that stays open.) $\mathcal{F}$ does not prove that the machine works; that is the external part. $\mathcal{F}$ has only the axiom.

The encoding enters as a floor. The horizon is measured in Gödel numbers, so the argument needs every symbol to cost bits: Eq. (20) requires on average more than five bits per U-Grounding symbol along the deepest branch, and 2001's Case 11 needs the sentence introducing a sum to require "substantially more than six bits" (2001, p. 28). Larger Gödel numbers only widen the

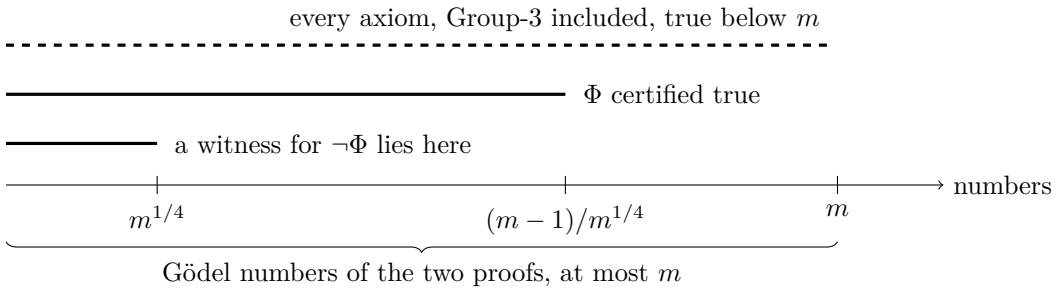


Figure 1: The two horizons in the proof of Willard's 2005 Theorem 1 (schematic, not to scale). A universal theorem is certified out to its horizon, an existential one finds its witness inside its horizon, and when horizons are short compared with the proofs, the two collide in the region where every axiom is true.

margin, and Willard says so: "All our theorems would also be valid for uncompressed encoding methods. However, they are more meaningful when one uses an efficiently compressed B-adic encoding method" (2005, p. 40, text layer; 2006a makes the same remark for the Hilbert line). What would break the 2005 proof of Theorem 2 is a notation *more* compact than Eq. (20) allows, with fewer than four bits per symbol on average, so that the U-Height bound of its third arrow fails. Whether Theorem 3 itself fails for such an encoding Willard does not say, and we do not know.

The closing inequality. Willard closes the proof of Theorem 1 with "since $\theta < \frac{1}{3}$ implies $g > h$", where $g = (m-1)/m^{\theta}$ and $h = m^{\theta}$ (p. 20). That needs $m - 1 > m^{2\theta}$, which holds for every $\theta < \frac{1}{3}$ once $m \geq 4$, and for $\theta = \frac{1}{4}$ once $m \geq 3$. Under Appendix A's encoding every proof has Gödel number above $32 \cdot 64^2$, so the gap is harmless; Theorems 3 and 5, which are corollaries of Theorem 1, inherit both the gap and its repair.

7 The Faith Is Not Just Missing

One might think Willard's systems are self-justifying merely because nobody has managed to prove step (2) in them. In fact they cannot prove it, and the reason is Oberhoff's Theorem 6 run backwards.

Proposition K (ours)**.** *Let $(\mathcal{F}, D)$ be a configuration with D $\mathcal{F}$-sound and $\mathcal{F}$-complete and $\mathrm{Prf}_{\mathcal{F}}^{D}$ bounded, and suppose:*

(H1) $\mathcal{F}$ is consistent;

(H2) $(\mathcal{F}, D)$ watches;

(H3) the hypotheses of Fact B hold, so (FP) holds;

(H4) $\mathcal{F}$ proves its own Level(1) consistency for $\mathrm{Prf}_{\mathcal{F}}^{D}$: $\mathcal{F} \vdash_D \forall x\, \forall y\, \forall p\, \forall q\, \neg[\mathrm{Pair}(x,y) \wedge \mathrm{Prf}_{\mathcal{F}}^{D}(x,p) \wedge \mathrm{Prf}_{\mathcal{F}}^{D}(y,q)]$.

Then:

(a) $\mathcal{F} \vdash_D (2) \leftrightarrow \ddot{\mathsf{G}}$ and $\mathcal{F} \vdash_D [\neg\ddot{\mathsf{G}} \supset \mathrm{Prov}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner)] \leftrightarrow \ddot{\mathsf{G}}$;

(b) $\mathcal{F} \nvdash_D \ddot{\mathsf{G}}$;

(c) hence $\mathcal{F}$ proves neither step (2) nor the instance of knowing that it watches at $\neg\ddot{\mathsf{G}}$, and both are true.

Proof. (a) Assume step (2). From (H4) and the true bounded sentence $\mathrm{Pair}(\bar{c}, \ulcorner\neg\ddot{\mathsf{G}}\urcorner)$, which $\mathcal{F}$ proves by (H2), $\mathcal{F}$ proves step (3): $\neg[\mathrm{Prov}(\bar{c}) \wedge \mathrm{Prov}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner)]$. With step (2) this gives $\neg\mathrm{Prov}(\bar{c})$, and (FP) gives $\ddot{\mathsf{G}}$. Conversely $\ddot{\mathsf{G}}$ gives $\neg\mathrm{Prov}(\bar{c})$ by (FP), which makes step (2) true. Fact A turns both directions into D-proofs. The second equivalence follows by Lemma E(b).

(b) This is Oberhoff's §4, case 1:

$\Rightarrow$ Suppose $\mathcal{F} \vdash_D \ddot{\mathsf{G}}$, by a proof m.

$\Rightarrow$ $\mathrm{Prf}^D_{\mathcal{F}}(\bar{c}, \bar{m})$ is a true bounded sentence, so $\mathcal{F} \vdash_D \mathrm{Prov}(\bar{c})$ by (H2): $\mathcal{F}$ watches $\mathsf{P}(\mathsf{P})$ halt.

$\Rightarrow$ By (FP), $\mathcal{F} \vdash_D \neg\ddot{\mathsf{G}}$, against (H1). $\lightning$

(c) By (a) and (b). Truth: since no proof of $\ddot{\mathsf{G}}$ exists, $\ddot{\mathsf{G}}$ is true in the standard model ($\mathrm{Prf}^D_{\mathcal{F}}$ and SUBST being correct), step (2) is true because its antecedent is false, and the instance is true because $\neg\ddot{\mathsf{G}}$ is false. □

In a system that proves its own Level(1) consistency, then, the faith step *is* the Gödel sentence, up to provable equivalence, and is true and unprovable for the same reason.

Corollary L (ours)**.** *Let A prove every Π_1^* theorem of* PA+*, let every Π_1^* theorem of A be true, and let D be semantic tableaux or* Tab$-U_1^*-$List*. Take $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}$ to be Willard's SubstPrf$^D_{\mathrm{UNION}(A)}(\bar{n}_\Gamma, \cdot, \cdot)$ and $\ddot{\mathsf{G}}$ built from it. Then $\mathrm{IS}_D(A)$ proves neither step (2) nor the instance of knowing that it watches at $\neg\ddot{\mathsf{G}}$.*

Proof. (H1) is Theorem H(a). (H2) is Lemma I. For (H3): Groups 0 and 1 prove the true bounded sentences; the order properties and (U) are Π_1^* theorems of PA+, inherited through Group-2; (E) holds for Willard's encoding (§2). (H4) is the Group-3 axiom, because $\ddot{\mathsf{G}}$ is built from the same proof predicate. □

One point in this corollary is a transposition and is ours: Willard asserts (U) for Δ_0^- substitution and PAX (2001, p. 39), and we use it for the Δ_0^* substitution formula of 2005 (p. 32) and PA+.

Corollary M (ours)**.** *Let $A \supseteq$ PAX be regularly consistent. Then $\mathrm{IS}(A)$ does not prove both step (2) and the Level(0−) form of step (3), $\forall p \forall q\,[\mathrm{Prf}(\bar{c}, p) \wedge \mathrm{Prf}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner, q) \supset \exists r\,\mathrm{Prf}(\ulcorner 0 = 1\urcorner, r)]$.*

Proof. If it proved both, then with its Group-3 axiom it would prove step (3) for Level(1) at the one pair $(\ddot{\mathsf{G}}, \neg\ddot{\mathsf{G}})$, which is all that the proof of Proposition K(a) uses of (H4). The other hypotheses hold as in Corollary L, now inside the setting of Willard's own Theorem 7.2 (Δ_0^- proof predicate, tableaux, the Π_1^- theorems of PAX inherited). Then $\mathrm{IS}(A) \vdash \ddot{\mathsf{G}}$, against Proposition K(b). □

So for the 2001 system the faith fails in at least one of two places, and we do not know which.

The proposition is a contrapositive and claims no novelty of method. Its sibling in Willard is 2001 Theorem 7.2 (p. 38, `full`): no consistent α with a Δ_0^- proof predicate, an α-sound and α-complete D, and all Π_1^- theorems of PAX can prove the canonical reflection principle $\forall y\,\{\mathrm{Prf}^D_\alpha(\ulcorner\Upsilon\urcorner, y) \supset \Upsilon\}$ for every Π_1^- sentence Υ. Its proof uses the same fixed point and the same Fact A in place of the derivability conditions; "The above result is different from Löb's Theorem … chiefly because we do not assume that α recognizes either Addition or Multiplication as total functions" (p. 38, text layer). Willard's 1994 framing already sets out the alternatives: "every axiom system either fails to prove its consistency or fails to meet the full Hilbert-Bernays conditions" (TR 93-10, p. 6, verbatim). What the computational reading adds is *which* condition fails, at *which* sentence, and that the failing instance is the Gödel sentence in disguise. What Willard adds, and no contrapositive can supply, is that systems on the far side of the alternative *exist* (Theorem H).

8 The Faith, Bought: Willard's Theorem 6

Willard's 2005 paper ends by doing the experiment. Take the transcript statement of §4, bound it, and see what it costs to switch it on.

Let Log^k be the k-fold base-2 logarithm, rounded down, and $\Upsilon(k,m)$ the true Σ_1^* sentence $\exists z\, \mathrm{Log}^k(z) \geq \bar{m}$, which says that a number at least 2_k^m exists, where 2_k^m, the least z with $\mathrm{Log}^k(z) \geq m$, is roughly a tower of k exponentials topped by m (2005, Definition 2, p. 11). $\mathrm{NS}_D^{k,m}(A)$ is $\mathrm{IS}_D(A)$ with $\Upsilon(k,m)$ added to Group-0 and Group-3's "this" adjusted to the larger system (Definition 3, p. 11). Both systems prove $\Upsilon(k,m)$; only one of them has it as an axiom. Let $\mho(s,b)$ be the bounded Gödel sentence "no proof p of me with $\mathrm{Log}^s(p) < b$" (Eq. 24, p. 33), and *CheckProof*(t,p,q) the Δ_0^* formula "p proves t, and q proves $\exists p\, \mathrm{Prf}(t,p)$" (Definition 6, p. 33). Lemma 4 (p. 33, `full`, resting on Lemma 3, a proof `sketch` that checking a proof is quasi-polynomially cheap) says that for k large Peano Arithmetic proves the Π_1^* sentence

$$\forall p \forall n \exists q < n \{[\mathrm{Log}^{k-1} p < m \wedge \mathrm{Log}^k n \geq m \wedge \mathrm{Prf}(\ulcorner \mho(k-1,m) \urcorner, p)] \\ \rightarrow \mathit{CheckProof}(\ulcorner \mho(k-1,m) \urcorner, p, q)\}. \quad (27)$$

This is step (2), bounded. For every short proof p of the Gödel sentence and every number n that is big enough, a transcript q exists below n. Because the transcript is bounded by n, the statement is Π_1^*, and the system inherits it from Peano Arithmetic through Group-2. What is missing is n. (Willard states Lemma 4 for the proof predicate of $\mathrm{NS}_D^{k,m}(\mathrm{PA}+)$. Nothing in the sketch of Lemma 3 is specific to the added axiom, so we expect the same bounded faith step for $\mathrm{IS}_D(\mathrm{PA}+)$'s own proof predicate, but Willard does not state it.)

Theorem N (Willard 2005, Theorem 6, p. 34, `full`)**.** *For* $D = \mathrm{Tab}{-}U_1^*{-}\mathrm{List}$ *there is* k_0 *such that for every* $k > k_0$, $\mathrm{NS}_D^{k,m}(\mathrm{PA}+)$ *is inconsistent.*

Willard's proof is a seven-step tableau list W (pp. 34–36), and it is Oberhoff's Theorem 6 with the faith supplied:

- $\Rightarrow$ (27), inherited from Peano Arithmetic: the faith step, below n.
- $\Rightarrow$ (28) $\mathrm{Pair}(\ulcorner \mho \urcorner, \ulcorner \neg \mho \urcorner)$, a true bounded sentence.
- $\Rightarrow$ (29) the Group-3 axiom, and (30) its instance: step (3).
- $\Rightarrow$ (31)–(32) chaining: no short proof of $\mho$ coexists with a big n.
- $\Rightarrow$ (33) $\Upsilon(k,m)$, an axiom, supplies n: no short proof of $\mho$ exists. This is $\mho(k-1,m)$ itself, which the seventh step derives.
- $\Rightarrow$ Because $\Upsilon(k,m)$ cost nothing, W is short: for $k > k_0$ its Gödel number is below 2_{k-1}^m, so W is itself one of the short proofs that $\mho(k-1,m)$ says do not exist. The system watches W and refutes $\mho$. ↯

Willard calls the length estimate "quite routine" (p. 36). He measures the difference between the two systems by a constant: proof lengths in $\mathrm{NS}_D^{k,m}(A)$ and $\mathrm{IS}_D(A)$ differ by at most some $C > 2_k^m$, "an enormous quantity" (p. 31). Why the argument does not transfer to $\mathrm{IS}_D(A)$ can be seen with Willard's own engine (our observation): $\mathrm{IS}_D(\mathrm{PA}+)$ must prove $\Upsilon(k,m)$ as a lemma, and since all its axioms are true Π_1^* sentences it is $\mathrm{Normed}(a,1)$ for every a, so by θ-compactification any proof p of $\Upsilon(k,m)$ has a witness below $p^{1/4}$. The least witness is 2_k^m, so $p \geq (2_k^m)^4$, and a tableau list containing p lies far beyond the bound 2_{k-1}^m of which $\mho(k-1,m)$ speaks. "A Kleene-like Group-3 axiom has a

different connotation when it refers to a short proof, referencing itself, than when it refers to a long such proof" (p. 32, text layer). The asymmetry is exact: adding a true Π_1^* sentence instead of the Σ_1^* sentence $\Upsilon(k, m)$ preserves consistency (p. 37, stated as a generalization of Theorem 5). A universal axiom constrains; an existential axiom supplies a number, and the number is what the faith step was missing. Remark 8 (p. 36, `stated-only`) extends the theorem to semantic tableaux and other cut-free methods, to empty A, and to systems that drop the totality of addition altogether.

9 The Price of Faith

If withholding faith is what lets $\ddot{\mathsf{W}}$ survive, the second incompleteness theorem should return as soon as the system can afford the transcripts. Willard spent the second half of his career proving that it does. The price depends on the language and on how much the proof system may reuse.

Squaring

Give a tableau system multiplication as a total function. Then a proof can introduce $u_0 = 2$, $u_{i+1} = u_i \cdot u_i$, and so reach $u_n = 2^{2^n}$, in $O(n)$ steps. The 2001 system with multiplication added, ISMULT(A), "satisfies an analogue of" Assertion ++ but has no analogue of Lemma 4.2, for exactly this reason (2001, Remark 4.5, p. 21). On the negative side Willard located the failure precisely (2000, §5, quoted in the codified statement §5.3):

> Many readers will smile with amusement when they learn the true reason that the Semantic Tableaux version of the Second Incompleteness Theorem breaks down when Multiplication is changed from a total function into a 3-way relation. It is essentially that Lemma 8 and its short proof then become no longer valid. ... Without it, our proof of the Semantic Tableaux version of the Second Incompleteness Theorem collapses entirely!

Lemma 8 (2000, p. 11, `full`) is the squaring construction: a tableau proof from $Q + V$, of length $O([\log \log z]^C)$, that a number about as large as z exists. With it, the second theorem comes back, at Level(0−):

- no consistent *finite* extension of $Q + V$, for an explicit Π_1 sentence V with $Q + V$ consistent, proves its own semantic tableaux consistency (2002a, Theorems 3.4–3.5, p. 11, `full`);
- every consistent extension of T-1 or T-2, the textbook and Wilkie–Paris axiomatizations of $I\Sigma_0$, is unable to prove its own semantic tableaux consistency (2007, Theorem 6, p. 29, `full`);
- any consistent extension of $I\Sigma_0$ with the Conventional Deciphering Property (a finite subset proves, of each axiom's canonical numeral, that it is an axiom: 2002a, Definition 5.1, p. 20) cannot prove that there is no tableau proof of $0 = 1$ from itself (2002a, Theorem 6.4, p. 25, `sketch`).

In 1981 Paris and Wilkie asked whether $I\Sigma_0$ satisfies the second incompleteness theorem for cut-free deduction. 2002a presented Theorem 6.4 as an answer; 2007 records that the 2002 paper covered only the textbook induction scheme ("which we did not examine in our year-2002 article", of the Wilkie–Paris scheme: 2007, abstract, text layer), and its Theorem 6 covers both. On the positive side, Definition 5 becomes "impossible to satisfy" once multiplication is admitted as a ninth function symbol (2005, Remark 4, p. 21, `stated-only` as to the reason: "intuitively because multiplication has a faster growth property than addition").[1]

[1] What multiplication destroys is the flat consistency axiom and canonical reflection, not every self-referential construction. Willard's ISM(A) recognizes multiplication as total and is consistent whenever A is regularly consistent,

Subroutines

A Hilbert-style proof can prove a lemma once and use it many times. Modus ponens is a subroutine call. A tableau cannot do this; every use must be inlined. Inlining can be astronomically expensive: by Gentzen, the tableau proof exists, but "t's length can certainly be super-exponentially longer" than the proofs it combines (2002a, Theorem 2.2, p. 7, `sketch`), and Boolos gave a famous small example of the blowup [Boolos 1984]. With subroutines, a little growth is amplified into a lot. Willard saw this in 1994 (TR 93-10, p. 7, verbatim):

> These metaphors lose meaning when cuts are present because Addition can then be employed to *implicitly define* a sequence that grows actually faster than $a_0, a_1, a_2, a_3 \ldots$. Solovay's proof that IS(A) is incompatible with deductive cuts [So94] rests on this fast growth property. (Actually, IS(A) can support a cut rule for Δ_0 sentences, and Solovay's theorem relies on the cut rule for sentences with unbounded quantifiers.)

The boundary depends on the *complexity* of the lemmas, and Willard mapped it:

- a cut rule for bounded (Δ_0) sentences: compatible with IS(A), as the parenthesis above says (TR 93-10, p. 7; the codification records the construction in a footnote on p. 35, obligation O56);
- lemmas with one unbounded quantifier block (Π_1^* or Σ_1^*), addition total: self-justifying at Level(1) (2005, Theorem 5, `full`);
- lemmas with two blocks (Π_2^*, respectively Σ_2^*), addition total: for explicit Π_1^* theorems V_A, V_B of Peano Arithmetic, no consistent system in the U-Grounding language containing V_A (resp. V_B) proves its own Level(0−) consistency under Tab−Π_2^* (resp. Tab−Σ_2^*) lists (2004, Theorem 2, p. 8, `sketch`; stated for finite systems "for the sake of simplifying our proof");
- unrestricted lemmas (Hilbert deduction), successor total together with the Π_1 algebra of $+$ and $\times$: Solovay's theorem below.

The unrestricted case runs through the classical shortening of definable cuts. In outline (a textbook gloss, not Willard's text): with lemmas, one defines a set of numbers and proves once that it is closed under successor; then, by a short proof reusing that lemma, that a smaller set is closed under addition, and then that a smaller set still is closed under multiplication. On the last set the system can afford transcripts, and Gödel's argument runs there. Pudlák showed that a consistent extension of Robinson's Q proves neither its Hilbert consistency nor its consistency restricted to any definable cut [Pudlák 1985] (Willard 2006a, Theorem 1, p. 4, `cited`). Solovay's theorem, communicated to Willard by telephone in April 1994 and never published by Solovay, states the conclusion (2006a, Theorem 2, p. 4, `cited`): a consistent system that formalizes arithmetic with three-place predicates for $+$ and $\times$, proves successor total, and proves the Π_1 associative, commutative, distributive and identity laws for those predicates cannot prove that there is no Hilbert proof of $0 = 1$ from itself. With unrestricted subroutines, the price of faith is mere counting.

So the Hilbert-line self-justifying systems give up successor itself. They name numbers with constants $C_1, C_2, \ldots$, and the naming convention *is* the system (codified statement §5.5). Incremental naming $C_i = C_{i-1} + 1$ is how 2006a reads 2001's ISREF(A), which is consistent when A is regularly consistent (2001, Theorem 3.4, p. 13, `full`) and whose Group-3 is the tangibility schema (2001, Eq. 10), not a flat consistency axiom: "The term 'incremental naming convention' had not appeared in our article [46]. However, its Theorem 3.4 had implicitly presumed the presence of an incremental

"by a routine generalization" of his proof, but its self-referential axiom is weaker, and it cannot prove the flat statement that no tableau proof of $0 = 1$ exists (2001, §6, p. 36; codified statement §7.2).

naming method" (2006a, p. 22, text layer). Additive naming $C_i = C_{i-1} + C_{i-1}$ works and keeps reaching new numbers (ISCE(A), for consistent A with true Π_1^- theorems and Δ_0^- Hilbert proof predicate: 2006a, Theorem 3, p. 6, `full`); multiplicative naming $C_i = C_{i-1} \cdot C_{i-1}$ fails for any system that contains the convention's axioms, proves Peano Arithmetic's Π_1^- theorems and has Willard's Concise Encoding property (2006a, Theorem 4, p. 7, `full`). Remark 3, quoted in §4, explains why.

Other ways to buy faith

Excluded middle as a logical axiom, $\Phi \vee \neg\Phi$ at any node for any Φ (Xtab, 2020, p. 11), is what TR 93-10 had already defined as "a tableaux proof with cuts" (printed p. 33, as recorded in codified statement §7.3). Then $\mathrm{IS}_{\mathrm{Xtab}}(\beta)$ is automatically inconsistent whenever β proves the usual Π_1^* associative, commutative, distributive and identity laws for addition and multiplication (2020, Theorem 4.5, p. 13, `sketch`). And letting $\ddot{\mathrm{W}}$ say more about itself brings the second theorem back: there is a Π_1^* theorem W of Peano Arithmetic such that no consistent finite system in the U-Grounding language containing W can prove its own Level(2^+) tableaux consistency, the absence of proofs of a Boolean combination of Π_2^* and Σ_2^* sentences together with its negation (2004, Theorem 1, p. 6, `full`, although the codification records that one step is deferred, in a footnote saying it "should not be considered a formal proof": codified statement §7.5).

Proof system	Growth	Consistency asserted	Outcome	Source, status
tableaux, or lists of Π_1^*/Σ_1^* lemmas	+ total, × a relation	Level(1)	self-justifying	2005 Thms 3, 5, `full`
same	same, plus one axiom $\exists z\,\mathrm{Log}^k z \geq m$	Level(1)	inconsistent	2005 Thm 6, `full` (Lemma 3 `sketch`)
tableaux	× total	Level(0−)	G2	2002a Thm 3.5 `full`; 6.4 `sketch`; 2007 Thm 6 `full`
lists of Π_2^* or Σ_2^* lemmas	+ total	Level(0−)	G2	2004 Thm 2, `sketch`
tableaux	+ total	Level(2^+)	G2	2004 Thm 1, `full` (one deferred step)
tableaux + excluded middle	+ total, × laws	Level(1)	inconsistent	2020 Thm 4.5, `sketch`
Hilbert	successor total, Π_1 laws of +, ×	Level(0−)	G2	Solovay, `cited`
Hilbert	additive names only	Level(0−)	self-justifying	2006a Thm 3, `full`
Hilbert	multiplicative names	Level(0−)	G2	2006a Thm 4, `full`

Read the table as Willard read it. Negative rows assert the weakest consistency notion and positive rows the strongest, by the rule of §2, so each row is as strong as its side allows. "Self-justifying" in a positive row means that the system proves its own consistency and is consistent under the hypothesis on its base theory (Theorem H; 2006a, Theorem 3). The affirmative results are mostly `full`; the negative side contains sketches and one unpublished theorem. The boundary is sharply located in some coordinates and only bracketed in others (codified statement §8.2).

10 Trust Within Reach

A system *trusts* its proofs if it proves canonical reflection, $\forall y\,\{\mathrm{Prf}_\alpha(\ulcorner\Psi\urcorner, y) \supset \Psi\}$, for every Π_1-class Ψ. That is too much to ask: 2001 Theorem 7.2 (§7) forbids it for any consistent system with a bounded proof predicate that inherits the Π_1^- theorems of PAX. What Willard's systems can prove is *tangibility reflection* (2001, Definition 1.2, p. 4, Eq. 2):

$$\forall x\ \{\ [\exists y\,\mathrm{Prf}_\alpha(\ulcorner\Psi\urcorner, y) \wedge \mathit{Tangible}(x)] \supset \Psi^x\ \},$$

where Ψ^x is Ψ with its unbounded universal quantifiers bounded by x, and *Tangible* is one of $\mathit{TangPred}(x) \equiv \exists v\, x < v - 1$, $\mathit{TangDiv}_k(x) \equiv \exists v\, x < v/k$, or $\mathit{TangRoot}_k(x) \equiv \exists v\, x < v^{1/k}$. ISREF($A$), a Hilbert-style system without addition total, proves the *TangPred* form for every prenex* Ψ, given that $\forall y\, \mathit{Size}(y) < y - 1$ (2001, Theorem 3.5, p. 14, `full`). The tableau system $\mathrm{IS}^\lambda(A)$ has a Hilbert proof of the $\mathit{TangRoot}_k$ form when $0 < \lambda < 1$, $k > 1/(1-\lambda)$, and A proves a stated Π_1^- sentence (Theorem 6.1, p. 34, `full`), and a tableau proof by cut elimination (Corollary 6.2, p. 35, `full`); its consistency is proved for $\lambda = 3/4$ (Theorem 5.1, p. 28, `full`), where the condition reads $k > 4$. In computational terms: *I trust my verdicts on every input for which there is memory to spare.* For a Σ_1-class claim the bound costs nothing, since $\Psi \equiv \Psi^0$; taking Ψ to be $0 = 1$ gives back the flat statement that no proof of $0 = 1$ exists (2001, p. 8).

11 Taking Off the Glasses

Oberhoff's §5 takes the glasses off $\ddot{\mathsf{G}}$ and finds "I am not provable" underneath. Take them off $\ddot{\mathsf{W}}$ and you find "nothing refutes me", that is, "I am consistent". Over Peano Arithmetic, $\ddot{\mathsf{W}}$ is refuted, and $\mathsf{C}(\mathsf{C})$ finds the refutation: Theorem 6, written out. In Willard's systems that refutation would have to establish the existence of a transcript-sized object from inside a proof smaller than the object. The snake cannot swallow its tail, because its mouth is smaller than its body.

So self-reference is not where the trouble lies. The fixed point builds $\ddot{\mathsf{W}}$ in Willard's systems as easily as in Peano Arithmetic. What costs something is *self-simulation*: running yourself in your head and writing down the result. The second incompleteness theorem needs both. The first needs only self-reference and watching, which is why Willard's systems obey it.

Oberhoff found that his classical and computational proofs were "the exact same". The same holds here. Willard's arithmetic (a least pair of contradictory proofs, values below $b \cdot 2^{\text{U-Depth}}$, five bits per symbol) and the computational picture (a shortest refutation, the memory a proof occupies, a walk down the tree) are one argument. The arithmetic of Gödel numbers is the bookkeeping of memory.

12 Refactoring the Incompleteness Theorems, Again

Oberhoff's Theorem 7 is two theorems. Stated as properties of a configuration (α, D), they are:

> **7a** (Watching, Definition C). $\alpha \vdash_D \sigma$ *for every true* Σ_1*-class sentence* σ*; in particular, for the halting sentence of each halting computation.*
>
> **7b** (Knowing that one watches, Definition D). $\alpha \vdash_D \sigma \supset \mathrm{Prov}_\alpha^D(\ulcorner\sigma\urcorner)$ *for every* Σ_1*-class sentence* σ*, as a schema.*

Oberhoff's Theorem 7, read as 7a, holds for every configuration in this essay: for extensions of Robinson's Q with Hilbert deduction (standard), and for Willard's systems (Lemma I). Read

as 7b, it holds for recursively axiomatized extensions of $I\Sigma_1$ with their canonical provability predicate [Hájek–Pudlák 1993], and fails for Willard's systems. Property 7a is enough for the first incompleteness theorem in all of Oberhoff's forms. Property 7b is what Theorem 6 takes on faith, and it does not follow from 7a: Willard's systems satisfy 7a (Lemma I), prove their own Level(1) consistency, and fail 7b at one sentence, $\neg\ddot{\mathsf{G}}$, where the failing instance is provably equivalent to the Gödel sentence (Proposition K, Corollary L). By Lemma E(a) they satisfy 7b at every true Σ_1-class sentence, so the failure is confined to runs that do not happen. Willard's Theorem 6 (§8) shows the bounded form of 7b at the Gödel sentence to be a Π_1 theorem of Peano Arithmetic, inherited through Group-2; what the system lacks is one number large enough to switch it on, and granting that number as an axiom makes it inconsistent.

So Gödel's work splits into three parts, not two. The first is programming the integers (7a). The second is teaching the integers that they have been programmed (7b). The third is the victory laps. Willard's self-justifying systems live in the gap between the first and the second. The second incompleteness theorem is a victory lap only for a runner who has already paid for 7b, and the receipt, in Willard's accounting, reads: multiplication if your lemmas have at most one unbounded quantifier block, addition if they may have two (`sketch`), counting if they are unrestricted (`cited`), and, at the finest grain, one large number granted as an axiom.

Moral of the story: you may declare yourself a very stable genius, and Willard's theorems will vouch for you provided your teachers told you only truths, you cannot multiply, and you reuse only lemmas with one unbounded quantifier. Reuse lemmas with two, and (by a sketch of Willard's) addition is already too much; reuse any lemma you like, and (by Solovay's unpublished theorem) counting is too much; and you must never be told, as an axiom, that a big enough number exists.

13 What Self-Justification Does Not Buy

Willard was more careful about the limits of his results than most of his readers have been, and an honest retelling must carry his caveats.

- **Relative consistency.** Theorem H assumes that the base theory's Π_1-class theorems are true. Nothing here shows any system consistent outright; the systems are as trustworthy as their base. Willard: "if A = Peano Arithmetic, IS(A) can prove all the Π_1^- theorems of Peano Arithmetic, and its consistency will be as certain as that of Peano Arithmetic!" (2001, p. 11, text layer).

- **A reduced question.** Willard (2016, Remark 4.9, p. 13) splits Hilbert's question in two: can a system verify its own consistency in a robust sense, and "Can logic systems 'appreciate' their own consistency in some REDUCED sense, that is diluted but not fully immaterial?" (text layer). The answer to the first is no, by Gödel, Hilbert–Bernays and Pudlák–Solovay. Everything above answers the second.

- **A one-line proof.** A system that proves its consistency from its own axiom $\ddot{\mathsf{W}}$ has "essentially a 1-line proof" (2011, Remark 6.16, p. 28), which is instinctive faith rather than proof-justification. The substantive justification lies outside: the theorem that *Probe* defeats every refutation, proved in the metatheory under the hypothesis on A.

- **Almost cheating.** Willard raised the objection against himself: "Is it not almost cheating when an axiom system verifies its own consistency by using $\oplus$'s formalized 'I am consistent' axiom as an intermediate step, to verify its own consistency? After all, such a technique can verify its own consistency only in a technically purely legalistic sense" (2016, §8). His answer, *Platonic*

Stability, is that a thinking being needs no sophisticated justification of its own reasoning, only a formalism in which presuming its own consistency provably will not spin it into inconsistency. Its formal statement, Corollary 8.2 (p. 29), opens "Let us assume Conjecture 6.6 is correct (as we are almost 100 % certain it is)"; Conjecture 6.6 (p. 24) is `stated-only`.

- **The first theorem stands.** Willard's systems are incomplete; they cannot decide $\ddot{\mathsf{G}}$ or $\ddot{\mathsf{R}}$.
- **Interpretation versus source.** The vocabulary of watching and knowing, the program C as a reading of Willard's Group-3, the horizon picture, Lemmas E–J, Proposition K and its corollaries, and the reading of Willard's Theorem 6 as Oberhoff's Theorem 6 with the faith supplied are ours, not Willard's formulations. His own words for the mechanism are "adequate memory" (1994) and the comparison, in 2006a Remark 3, between a proof's Gödel number and the integer the proof shows to exist.

(The cake promised in Oberhoff's abstract is still on its way. We cannot prove that it exists: it is one multiplication away.)

A Provenance

Page numbers refer to the author copies in the repository's `nachlass/papers/`; statuses are those of `nachlass/codification/registry/results.md`. "Verbatim, image" marks quotations checked against rendered page images; "text layer" marks quotations checked only against the extracted text.

Paper	Item	Page	Content used here	Status
TR 93-10	—	6	philosopher's hedge; G1 but not G2; "either fails ... or fails to meet"; *Turing*(x, y, z)	verbatim, image
TR 93-10	—	7	adequate memory; a_{300}; cuts and Solovay	verbatim, image
TR 93-10	—	33	tableaux with $\Phi \vee \neg\Phi$ (Xtab)	via codified §7.3
2000	Lem 8; §5	11; —	squaring construction; "the true reason"	full; verbatim (codified §5.3)
2001	Def 1.2	4	tangibility predicates, Eq. 2	—
2001	§2	9–12	grounding functions; IS(A) groups; regular consistency	text layer
2001	Thm 3.4, 3.5	13–14	ISREF(A) consistency; *TangPred* reflection	full
2001	Lem 4.1, 4.2	17, 19	constants below $\sqrt{y}$; parameters below y	full; sketch
2001	Thm 4.3; Rem 4.4	20	IS(A) consistency; Assertion ++	full; verbatim, image
2001	Rem 4.5	21	ISMULT(A) has no analogue of Lem 4.2	—
2001	Lem 4.6, 4.8	22–23	affirmative branch cannot close; least refutation has one	full
2001	Case 11	28	"substantially more than six bits"	image (extraction record)
2001	Thm 5.1, 6.1; Cor 6.2	28, 34–35	IS$^{\lambda}(A)$ at $\lambda = 3/4$; *TangRoot*$_k$ reflection	full
2001	§6	36	ISM(A)	routine generalization (Willard)
2001	Lem 7.1; Thm 7.2, 7.3	37–40	closure; no canonical reflection, fixed point (63), (67); XIS$^{\lambda}$	full; text layer
2002a	Thm 2.2	7	cut elimination, super-exponential length	sketch
2002a	Thm 3.4, 3.5	11	$Q + V$: G2 for finite extensions, tableaux	full
2002a	Def 5.1; Thm 6.4	20, 25	Conventional Deciphering; $I\Sigma_0$	—; sketch
2002c	—	2	weakest notion for G2, strongest for exceptions	via obligation O59
2004	Thm 1, 2	6, 8	Level(2^+) G2; Π^*_2/Σ^*_2 lemmas	full (deferred step); sketch
2005	§1	—	Kleene, Rogers, Jeroslow	verbatim (codified §2.2)
2005	§2–3	4–10	U-Grounding language; Level notions; tableaux; Tab lists; IS$_D(A)$; Eqs. 5–8	text layer
2005	Def 4, 5; Thm 1	19–20	Normed, θ-compactification; engine	full
2005	Rem 2, 4	21	2001 proof hard to visualize; multiplication	—; stated-only
2005	Lem 1, 2; Thm 2	23–26	*Probe*; non-closure; $\theta = 1/4$; Eq. 20	full
2005	Thms 3–5	27–29	IS$_D(A)$ consistency-preserving	full
2005	Defs 2, 3, 6; Lem 3, 4; Thm 6	11, 33–36	NS$^{k,m}_D$; (27)–(33)	full (Lem 3 sketch)
2005	Rem 8; p. 37	36–37	stronger Thm 6; Π^*_1 additions	stated-only
2005	App A	39–40	encoding; (E); uncompressed encodings	text layer
2006a	Thms 1, 2	4	Pudlák's and Solovay's theorems	cited
2006a	Thms 3, 4	6–7	additive naming works; multiplicative fails	full
2006a	Rem 3	24	proof's Gödel number vs. the integer it proves exists	verbatim, image
2007	Thm 6	29	T-1, T-2: G2 for tableaux	full
2011	Rem 6.16	28	"essentially a 1-line proof"	—
2016	Def 3.4, Ex 3.5	7	self-justification; "ironically typically useless"	verbatim (codified §2.2)
2016	Rem 4.9; §8	13; —	the two questions; "almost cheating"	verbatim (codified §1.4)
2020	Thm 4.5	13	IS$_{\text{Xtab}}$ inconsistent	sketch

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